Dual-Detect: Cell characterization using mRNA AND microRNA

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1 Introduction

1.1 Motivation and Significance of the Research

Synthetic biology enables the design of programmable genetic circuits, offering precise control over gene expression and cellular behavior. Among the most versatile designs are logic gates that integrate multiple inputs to regulate transcriptional outputs in a Boolean manner. This project explores the design and implementation of a general-purpose two-input AND gate, where one input is a microRNA (miRNA) and the other is a messenger RNA (mRNA) species. Such logic circuits that include both miRNA and mRNA inputs are particularly valuable in therapeutic contexts, where precise cell-type or state-specific expression of genes is required, such as detecting Multiple Sclerosis (MS). Integrating post-transcriptional regulation via miRNAs with traditional mRNA-based transcriptional control offers new opportunities for complex cell programming, diagnostics, and targeted interventions.

1.2 Purpose of the Experiment

The primary goal of this study is to build and characterize a synthetic AND gate in which gene expression is contingent on the presence of both an mRNA input (IL6), which in this case is a proxy for CD31 mRNA which is a marker for endothelial cells, and the absence of a specific miRNA input (MT1), which in this case is a proxy for miRNA125 which is high in MS endothelial cells. The gate operates such that expression of an output reporter gene (mKO2) only occurs when ADAR-mediated RNA editing is triggered by IL6 mRNA, and CasE-mediated repression is blocked by MT1 miRNA. The high-level circuit is shown in Figure 1. We aim to quantify how well the full logic behavior is achieved by integrating these distinct molecular inputs and validate that the output is reliably restricted to conditions where both inputs are satisfied.

1.3 Conceptual Approach

Our approach centers on engineering a synthetic RNA-based AND gate that integrates two distinct molecular inputs—an mRNA (IL6) and a miRNA (MT1)—to precisely control gene

expression in mammalian cells. Specifically, we aim to develop a circuit that produces an output (mKO2) exclusively when both conditions are met: the presence of IL6 mRNA, indicative of endothelial cell identity, and the presence of MT1 miRNA, characteristic of a molecular signature associated with Multiple Sclerosis (MS) endothelial cells. To illustrate the real-world applicability of this system, we envision its deployment as a targeted diagnostic tool. For instance, it could be used to selectively identify and visualize endothelial cells implicated in MS pathology by detecting the simultaneous expression of cell-type-specific mRNA markers and disease-associated miRNAs. This targeted identification approach has significant therapeutic implications, allowing for precise interventions in diseased cells while minimizing impact on healthy tissue. As mentioned before, Figure 1 illustrates the conceptual mechanism of the Dual-Detect circuit clearly, highlighting the regulatory roles of IL6 mRNA and MT1 miRNA inputs. IL6 input triggers ADAR-mediated RNA editing to enable mKO2 reporter expression, while MT1 miRNA input inhibits CasE-mediated repression, thus ensuring that mKO2 is produced exclusively under dual-positive conditions. By experimentally validating this logic gate, we provide a foundational proof-of-concept for future designs of advanced RNA-based genetic circuits, potentially facilitating novel diagnostic and therapeutic strategies in synthetic biology and precision medicine.

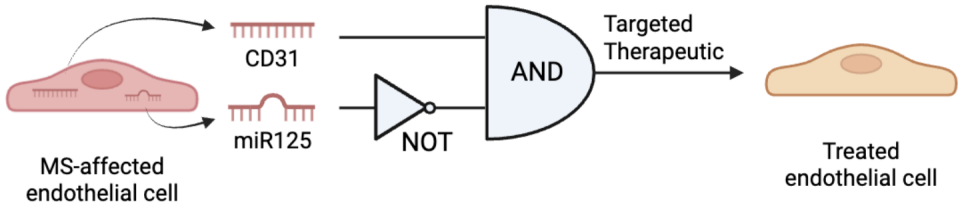


Figure 1: High-level circuit diagram showing how the Dual-Detect circuit can detect CD31 mRNA and miR125 miRNA specific to endothelial MS cells.

Note that there is an additional NOT operator on the lower miRNA branch. This is because the miRNA is intrinsically repressive while the mRNA is intrinsically promotive. Because we want the logic to output high when mRNA and miRNA are both high, the circuit needs an additional NOT operator to flip the logic for the miRNA.

1.4 Experimental Design

To implement this logic, the circuit is modularly split into submodules for stepwise testing. Constructs were assembled using Golden Gate Assembly and verified through restriction digestion. The experimental modules include: (1) testing CasE repression in the absence of MT1, (2) assessing IL6-dependent ADAR activation of the mKO2 reporter, (3) characterizing MT1-induced derepression of CasE via TagBFP output, and (4) full circuit integration to assess the complete AND logic. Constructs were transfected into CHO-K1 cells using Lipofectamine, and fluorescence was measured by flow cytometry. Computational models using Gro were also developed to simulate expected behaviors and guide the interpretation of experimental results.

1.5 Hypothesis and Predictions

Each of the four submodules tests a certain part of the circuit: CasE repression, MT1 repression, CasE and MT1 cascading repression, and IL6 ADAR repression system, respectively. We hypothesized that the submodels work as expected. High output signal for low CasE, high output at low MT1, high output at low CasE and high MT1, and high output at high IL6 and high ADAR, for the four submodules. Unfortunately, we were not able to test a full circuit condition, due to repeated failure to assemble and clone the actual output plasmid with CasE binding site and IL6 sensor. However, we hypothesize that output expression (mKO2) will occur at high dual-input conditions: (1) the presence of IL6 to activate ADAR-mediated editing and (2) the presence of MT1 to inhibit CasE repression. In all other cases, mKO2 output should be absent or significantly reduced. The summary of the actual findings is shown in the following section.

1.6 Summary of Findings and Implications

For the most part, the submodules all yielded results that show the expected trend. That being said, there were challenges with transfection under some conditions in the submodules that limit the clarity of the results. These limitations and the detailed findings are discussed in the Results section. For all experiments, we added transfection markers for the respective output plasmids. Across the experiments, the strength of the output directly correlates with the transfected output plasmids, as expected. Beyond that, the result saw high output signal for low CasE, high output at low MT1, high output at low CasE and high MT1, and high output at high IL6 and high ADAR, for the four submodules as hypothesized. As mentioned, the full circuit was not tested due to the failure to construct the full circuit output plasmid.

1.7 Circuit Diagram

A plasmid-level schematic of the full circuit is shown in Figure 2. The circuit integrates an mRNA input (IL6) and a miRNA input (MT1) to control output expression of mKO2. The IL6 mRNA activates ADAR, which edits the IL6 sensor to allow mKO2 expression. In parallel, MT1 inhibits CasE-mediated repression of the TagBFP reporter. Only in the presence of IL6 and MT1 is the full fluorescent output (mKO2) expressed. Fluorescent markers (neonGreen, iRFP720) serve as transfection controls.

2 Methods

2.1 Cloning

2.1.1 Plasmid Construction (Golden Gate Assembly)

Level 0 plasmids, such as backbones, promoters, and genes, were assembled into level 1 plasmids using Golden Gate Assembly with the NEBridge protocol (New England Biolabs). Table 1 and Table 2 in the Supplementary Materials section show a list of provided level 0 and 1 plasmids. Table 3 in the same section shows successfully cloned level 1 plasmids and

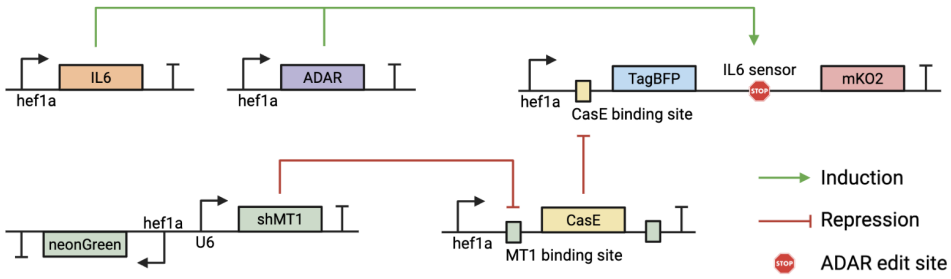


Figure 2: Plasmid-level schematic of the proposed AND gate circuit using IL6 and MT1, as proxy for general mRNA and miRNA species and mKO2 and proxy of detection signaling agents or therapeutics.

their level 0 constituents. The assembly uses 50 ng of each input plasmid and the Golden Gate Master Mix. E. coli 5-alpha was used for all plasmid construction and grown in Luria-Bertani (LB) medium (Miller, BD Biosciences) supplemented with kanamycin, ampicillin, or chloramphenicol (34ug/mL) as appropriate (Sigma).

2.1.2 MiniPrep

Bacterial cultures harboring the plasmid of interest were grown overnight in an LB medium supplemented with the relevant antibiotic. Plasmid DNA was isolated using the Monarch MiniPrep Kit (New England Biolabs). Finally, a Nanodrop machine was then used to calculate extracted DNA concentrations.

2.1.3 Transformation

Chemically competent E. coli 5-alpha cells were used for plasmid transformations: Thaw: Aliquots (10 µL each) of competent cells were thawed on ice for 5 minutes. DNA Addition: 1 µL of the plasmid or assembly product was gently mixed into 10 µL of cells. Heat Shock: The mixture was incubated at 42 °C for 45 seconds, then immediately placed on ice for 2 minutes. Recovery: 100 µL of SOC medium was added, and the culture was shaken at 37 °C, 280 rpm, for 60 minutes. Plating: 10 µL of the recovered culture was spread on LB plates containing antibiotics, followed by overnight incubation at 37 °C.

2.1.4 Restriction Digest and Gel Electrophoresis

Restriction digests and gel electrophoresis were performed to verify plasmid constructs. Plasmids were digested in relevant enzymes (determined by benching simulation) at 37 °C in 10X NEB CutSmart Buffer. The gel results then get confirmed against the expected band pattern from Benchling's in-silico tool.

2.2 Cell Culture

CHO-K1 cell lines were used. All cell lines used in this study were maintained in Dulbecco's modified Eagle medium (DMEM, Cellgro) supplemented with 10% FBS (Atlanta BIO), 1% penicillin/streptomycin/L-glutamine (Sigma-Aldrich), and 1% nonessential amino acids (11140050, ThermoFisher) at 37 °C.

2.3 Transfection

CHO-K1 cells were seeded in 24-well plates to reach 60-70% confluency. Transfections were performed with the Lipofectamine 3000 Kit (L3000001, ThermoFisher). We used OptiMEM, and DMEM as transfection culture media.

2.4 Computational Methods with Gro

Modeling assumptions for the Gro simulations are are shown below:

1. Unlimited transcription and translation resources
2. Protein creation is fast relative to protein degradation
3. DNA with the same promoters is expressed at similar rates
4. Co-transfected plasmids will be initialized at fixed ratios
5. Assuming cell plasmid uptake is normally distributed

2.4.1 Gro Simulation 1: Characterizing CasE Repression

In Gro Simulation 1, we will simulate CasE Repression on the production of mKO2. The measured species of interest are: TagBFP, neonGreen, and mKO2, which are the transfection markers for CasE, output plasmids, and the output signal, respectively. More details on the simulation are in the Results section.

Chemical reactions in Gro Simulation 1 are shown below:

Construction Reactions

$$\text{plasmid}_{\text{CasE}} \xrightarrow{k_{\text{translation}}} \text{CasE} + \text{plasmid}_{\text{CasE}}$$

$$\text{plasmid}_{\text{TagBFP}} \xrightarrow{k_{\text{translation}}} \text{TagBFP} + \text{plasmid}_{\text{TagBFP}}$$

$$\text{plasmid}_{\text{output}} \xrightarrow{k_{\text{transcription}}} \text{mRNA}_{\text{output}} + \text{plasmid}_{\text{output}}$$

$$\text{plasmid}_{\text{neonGreen}} \xrightarrow{k_{\text{translation}}} \text{neonGreen} + \text{plasmid}_{\text{neonGreen}}$$

CasE Association/Dissociation with mRNA

$$\text{mRNA}_{\text{output}} + \text{CasE} \xrightarrow{k_{\text{association}}} \text{mRNA}_{\text{output}}\cdot\text{CasE} + \text{mRNA}_{\text{output}}$$

Assuming CasE dissociation reaction is negligible.

Output Reactions

$$\text{mRNA}_{\text{output}} \xrightarrow{k_{\text{translation}}} \text{mRNA}_{\text{output}} + \text{mKO2}$$
$$\text{mRNA}_{\text{output}}\cdot\text{CasE} \xrightarrow{k_{\text{degradation}}} \varnothing$$

Unidirectional since cleavage is irreversible.

Degradation Reactions

$$\text{CasE} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{TagBFP} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{neonGreen} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{mKO2} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{mRNA}_{\text{output}} \xrightarrow{k_{\text{degradation}}} \varnothing$$

2.4.2 Gro Simulation 2: Characterizing MT1 Repression

In Gro Simulation 2, we will simulate MT1 Repression on the production of mKO2. Note that the MT1 repression mechanism differs from the CasE repression mechanism in that MT1 repression is modeled with a Hill function with the propensity of 1, while CasE repression was modeled to be linear. MT1 was also modeled to be reusable, while CasE was modeled to be single-use. The measured species of interest are: neonGreen, TagBFP, and mKO2, which are the transfection markers for MT1, output plasmids, and the output signal, respectively. More details on the simulation are in the Results section.

Chemical reactions in Gro Simulation 2 are shown below:

Construction Reactions

$$\text{plasmid}_{\text{MT1}} \xrightarrow{k_{\text{translation}}} \text{MT1} + \text{plasmid}_{\text{MT1}}$$
$$\text{plasmid}_{\text{neonGreen}} \xrightarrow{k_{\text{translation}}} \text{neonGreen} + \text{plasmid}_{\text{neonGreen}}$$

MT1 Association/Dissociation with mKO2 Plasmid

$$\text{plasmid}_{\text{mKO2_MT1_site}} + \text{MT1} \xrightarrow{k_{\text{association}}} \text{plasmid}_{\text{mKO2_MT1_site}}\cdot\text{MT1}$$
$$\text{plasmid}_{\text{mKO2_MT1_site}}\cdot\text{MT1} \xrightarrow{k_{\text{dissociation}}} \text{plasmid}_{\text{mKO2_MT1_site}} + \text{MT1}$$

Output Reactions

$$\text{plasmid}_{\text{mKO2_MT1_site}} \xrightarrow{k_{\text{translation}}} \text{plasmid}_{\text{mKO2_MT1_site}} + \text{mKO2}$$
$$\text{plasmid}_{\text{TagBFP}} \xrightarrow{k_{\text{translation}}} \text{TagBFP} + \text{plasmid}_{\text{TagBFP}}$$

Degradation Reactions

$$\text{MT1} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{neonGreen} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{TagBFP} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\text{mKO2} \xrightarrow{k_{\text{degradation}}} \varnothing$$

2.4.3 Gro Simulation 3: Characterizing IL6/ADAR Induction

In Gro Simulation 3, we will simulate IL6 and ADAR induction on the production of mKO2. The measured species of interest are: neonGreen, iRFP720, and mKO2, which are the transfection markers for IL6, ADAR, and the output signal, respectively. We perform these simulations at various values of transfected output plasmid, effectively abstracting away the need for a transfection marker for the output plasmid. More details on the simulation are in the Results section.

Chemical reactions in Gro Simulation 3 are shown below:

Construction Reactions

$$\text{plasmid}_{\text{IL6}} \xrightarrow{k_{\text{transcription}}} \text{plasmid}_{\text{IL6}} + \text{mRNA}_{\text{IL6}}$$
$$\text{plasmid}_{\text{neonGreen}} \xrightarrow{k_{\text{translation}}} \text{plasmid}_{\text{neonGreen}} + \text{neonGreen}$$
$$\text{plasmid}_{\text{ADAR}} \xrightarrow{k_{\text{translation}}} \text{plasmid}_{\text{ADAR}} + \text{ADAR}$$
$$\text{plasmid}_{\text{iRFP720}} \xrightarrow{k_{\text{translation}}} \text{plasmid}_{\text{iRFP720}} + \text{iRFP720}$$
$$\text{plasmid}_{\text{output}} \xrightarrow{k_{\text{transcription}}} \text{plasmid}_{\text{output}} + \text{mRNA}_{\text{output}}$$

IL6 Association/Dissociation with Output mRNA

$$\mathrm{mRNA_{output}} + \mathrm{mRNA_{IL6}} \xrightarrow{k_{\text{association}}} \mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}}$$
$$\mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}} \xrightarrow{k_{\text{dissociation}}} \mathrm{mRNA_{output}} + \mathrm{mRNA_{IL6}}$$

ADAR Association/Dissociation with mRNA Complex

$$\mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}} + \mathrm{ADAR} \xrightarrow{k_{\text{association}}} \mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}}\cdot\mathrm{ADAR}$$
$$\mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}}\cdot\mathrm{ADAR} \xrightarrow{k_{\text{dissociation}}} \mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}} + \mathrm{ADAR}$$

Editing of Output mRNA by ADAR

$$\mathrm{mRNA_{output}}\cdot\mathrm{mRNA_{IL6}}\cdot\mathrm{ADAR} \xrightarrow{k_{\text{editing}}} \mathrm{mRNA_{output\ (edited)}}$$

Note: Editing rate reflects IL6 effectiveness.

Output Reactions

$$\mathrm{mRNA_{output}} \xrightarrow{k_{\text{BFP}}} \mathrm{mRNA_{output}} + \mathrm{BFP}$$
$$\mathrm{mRNA_{output}} \xrightarrow{k_{\text{mKO2 (slow)}}} \mathrm{mRNA_{output}} + \mathrm{mKO2} \quad \textit{(slow)}$$
$$\mathrm{mRNA_{output\ (edited)}} \xrightarrow{k_{\text{BFP}}} \mathrm{mRNA_{output\ (edited)}} + \mathrm{BFP}$$
$$\mathrm{mRNA_{output\ (edited)}} \xrightarrow{k_{\text{mKO2 (fast)}}} \mathrm{mRNA_{output\ (edited)}} + \mathrm{mKO2} \quad \textit{(very fast)}$$

Note: $k_{\text{mKO2 (slow)}} \ll k_{\text{BFP}}$, $k_{\text{mKO2 (fast)}} \gg k_{\text{BFP}}$

Degradation Reactions

$$\mathrm{mRNA_{IL6}} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\mathrm{ADAR} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\mathrm{BFP} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\mathrm{neonGreen} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\mathrm{iRFP720} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\mathrm{mRNA_{output}} \xrightarrow{k_{\text{degradation}}} \varnothing$$
$$\mathrm{mRNA_{output\ (edited)}} \xrightarrow{k_{\text{degradation}}} \varnothing$$

2.5 FACS Data Collection and Analysis with Cytoflow

All fluorescence and scattering data were obtained using three different flow cytometry instruments over the course of the project: the SH800 (SONY) in the BioMakerspace, the Koch Fortessa HTS-2, and the Weiss Lab Fortessa. Each instrument was configured to detect side scatter (SSC), forward scatter (FSC), and relevant fluorescent channels based on the specific fluorescent proteins used in our genetic circuit. BioMakerspace SH800 (SONY): This machine was used to collect the majority of data, including preliminary and circuit validation experiments. Koch Fortessa HTS-2 color channels are as follow - (Blue: BV421), (Green: BB515), (Red: PE-Texas Red), (iRed: A700-A). Weiss Lab Fortessa color channels are as follow - (Blue: Pacific Blue-A), (Green: FITC-A), (Red: PE-Texas Red-A), (iRed: APC-Cy7-A). For all samples, cells were trypsinized (0.05% trypsin, 0.53 mM EDTA, Corning) 30 minutes prior to data collection and resuspended in 5PSGN medium to ensure a single-cell suspension prior to measurement.

The FACs data analysis workflow involved using Gaussian mixture model to morphologically gate and isolate single cells population, correcting for auto fluorescence baseline, correcting for spectral bleedthrough correction between overlapping channels, binning by transfection markers to normalize for variable plasmid copy number for co-transfection experiments, and finally, visualizing the outputs by fluorescence strength in heatmaps. All flow cytometry data analysis was performed using the Cytoflow Python package. A visual summary of the processing workflow is shown in Figure 3.

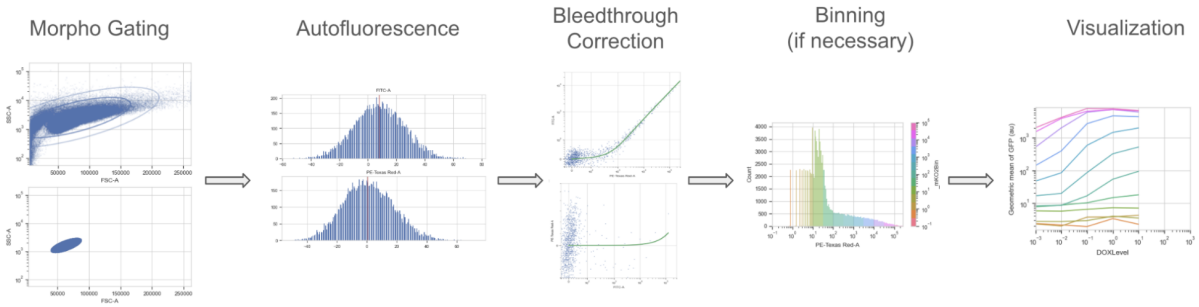


Figure 3: Visual representation of the FACs data analysis workflow.

2.6 Luciferase assay protocol

Luciferase assay measurement is performed for Transfection D. The protocol for the measurement is shown in Figure 23 in the Supplementary Materials section.

3 Results

As mentioned in the Methods section, the project is divided into four transfection submodules (A, B, C, and D) and three Gro simulations that follow Transfection A, B, and

D. Each transfection experiment is designed to characterize a different part of the circuit. A summary of relevant species and their functional descriptions is shown in Table 4 in the Supplementary Materials section.

3.1 Transfection A - Characterizing CasE Repression

The first component to characterize is the CasE repression mechanism. Transfection A is designed for that. Figure 4 shows the plasmid-level circuit diagram for Transfection A. Green is the transfection marker for the output plasmid that CasE targets, blue is the marker for the CasE plasmid, and red is the output. Figure 5 shows the transfection efficiency for the four samples: *Full poly*, *Full Co*, *Negative CasE Poly*, and *Negative CasE Co*. The Full samples test when both mKO2 and its repression target are presented. The Negative CasE samples test when only the repression target is presented and not CasE. Note that the Negative CasE sample still gets transfected with TagBFP to allow results to be viewed as a heatmap. It is expected that the Negative CasE samples will have higher output than the Full samples due to a lack of CasE to repress the mKO2 output. Transfection A also wants to look at the effect of Co and poly-transfection on the result.

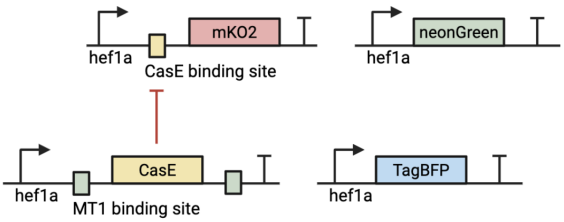


Figure 4: Plasmid-level circuit diagram for Transfection A testing the repression mechanism of CasE on mKO2 output plasmid with CasE binding site.

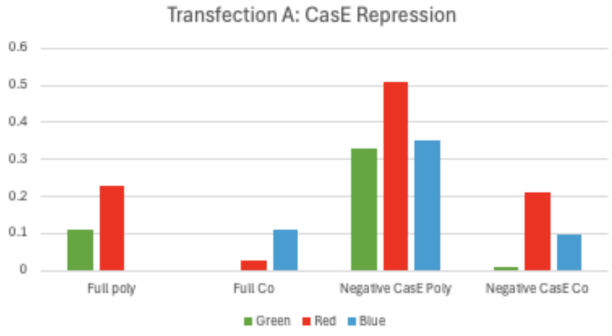


Figure 5: Transfection Efficiency for Transfection A for *Full Poly*, *Full Co*, *Negative CasE Poly*, and *Negative CasE Co* samples.

That being said, transfecting plasmids did not go completely according to plan for Transfection A. As seen in Figure 5, *Full poly* did not exhibit blue fluorescence. This means that the TagBFP plasmids, which were transfected separately with CasE in Transfection A, were not

transfected properly into the samples. This will result in a premature cutoff of datapoints on the x-axis, as will be seen in Figure 8. We also saw no green fluorescence for the Full Co and a negligible amount for the Negative CasE Co samples. This means that the mKO2 output plasmids were not transfected well into the two samples, which resulted in a lower output signal than expected, and generally, results that don't give much information.

3.1.1 Gro Simulation 1: Characterizing CasE Repression

Gro simulation 1 seeks to confirm the previously stated hypothesis that output expression is high at high copy number and low CasE presence.

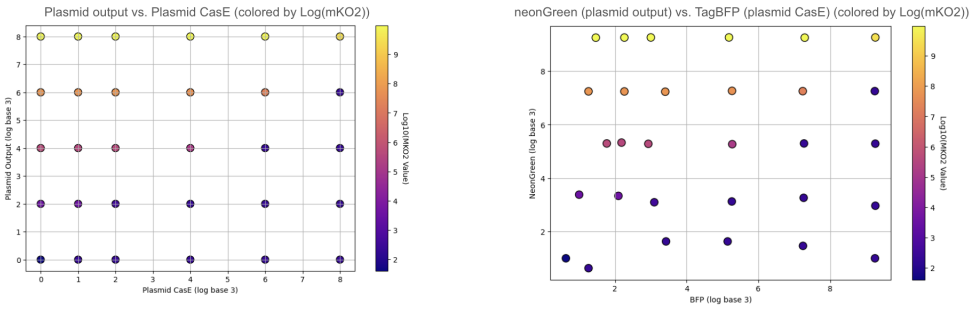


Figure 6: Heatmaps from Gro Simulation 1 to be compared with *Full Poly* sample from Transfection A. Left and right subplots show output signal by direct number of plasmids and fluorescence value of transfection markers, respectively.

Figure 6 shows the result from the simulation. The left and right plots show the output signal as a function of output plasmids vs CasE and neonGreen vs TagBFP fluorescence, respectively. Given that in real experiments, we measure fluorescence and not directly the plasmid counts, the right subfigure is a closer representation of expected results from real experiments. According to the constructed heatmaps, the hypothesis was confirmed. The output signal is highest at high green and low blue values, corresponding to high output plasmid copy number and low CasE presence. We expect CasE to work, and the Transfection A results to qualitatively follow the same trends in simulation 1. Transfection A results show generally similar results to simulation 1, but due to issues with transfection, the results did not completely coincide. This is further discussed in the following section. Also, it is worth noting that simulation 1 and subsequent simulations are set up to mimic poly-transfection plasmid uptake. This is because this project is interested in understanding the dose-response. Justification for poly-transfection over co-transfection was tested in Transfection A and discussed in the following section on the *Full* sample.

3.1.2 Transfection A Results

As seen in both Figure 7 and 8, poly transfection yields a higher spread of Green and Blue values in the heatmap. This is expected, as poly transfection allows different cells to uptake different proportions of the plasmids compared to co-transfection. For the Negative samples

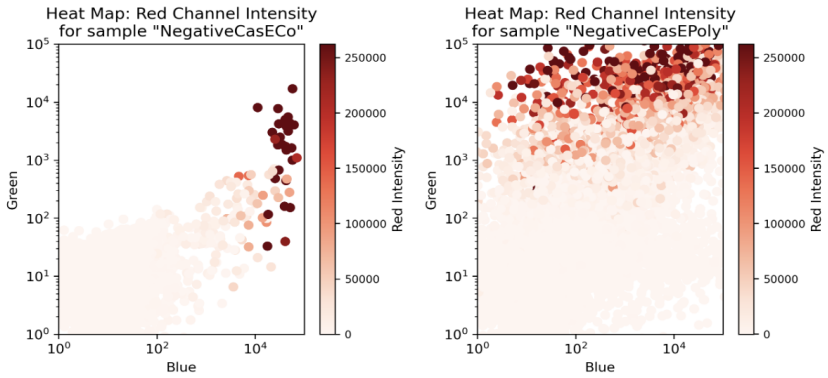


Figure 7: Heatmaps for the *Negative CasE Poly* and *Negative CasE Co* in Transfection A.

in Figure 7, we expected no correlation between the red output and the blue fluorescence values, as the samples were not introduced with CasE plasmids and just the TagBFP marker. We also expected a positive correlation with the y-axis, or the amount of output plasmids transfected. Negative CasE Poly behaved as expected. Negative CasE Co., on the other hand, exhibits an unexpected result. First, there's a relatively smaller red signal overall. Second, there seems to be a second population of cells at high values of blue that seems to exhibit high output. A possible explanation for the weird distribution can be that Negative CasE Co was not transfected well with the output plasmid as previously discussed, and co-transfection resulted in a non-uniform plasmid uptake that results in pockets of uptake ratio that are reflected in the heatmap. If one were to impute cells with high green uptake and low blue uptake into the map, one would expect to see cells with similar values to those of the ones with high green and high blue in the map.

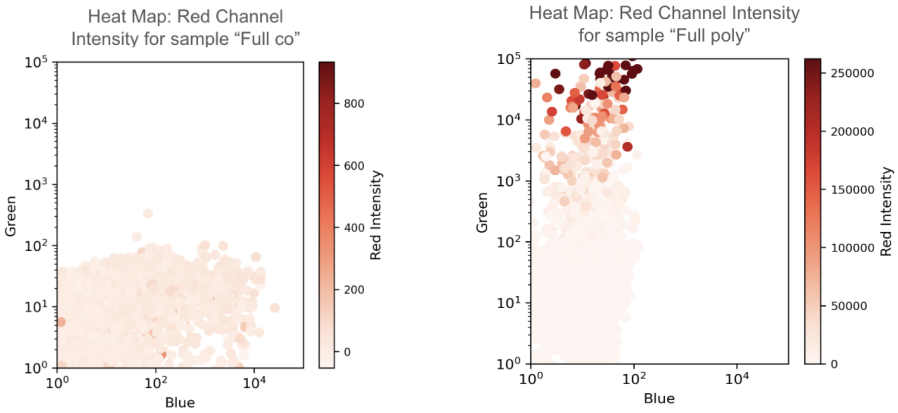


Figure 8: Heatmaps for the *Full Co* and *Full Poly* in Transfection A.

According to Figure 5, the Full Co sample did not have any green fluorescence, which means it is very likely that the sample was not transfected with the output plasmid at all. The

sample also has low red fluorescence, which supports this hypothesis. This could explain the lack of red signal in the Full Co sample. The Full Poly sample has a small spread of blue values, which, as mentioned, could be a result of poor blue plasmid transfection. Had blue been transfected well into the Full Poly sample, we expected to see a general drop in output as the blue value increases, a proxy for higher CasE amount in the sample. We repeated this exact condition later in Transfection C and confirmed the behavior. To summarize the results from Transfection A, poly transfection gives a higher spread of plasmid uptake ratio, and CasE repression works as expected at various output plasmid concentrations. Because this project is interested in the dosage response of the system, poly-transfection offers more spread in data points, and so the following submodules were designed around poly-transfection.

3.2 Transfection B - Characterizing MT1 Repression

Following the characterization of the mRNA CasE repression in Transfection A, Transfection B is designed to characterize how the MT1 miRNA repression works. Figure 9 shows the plasmid-level circuit diagram for Transfection B. For Transfection B, blue is the transfection marker for the output plasmid that MT1 targets, green is the marker for the MT1 plasmid, and red is the output. Figure 10 shows the transfection efficiency for the five samples: *Full-1* and *Full-2*, which are just replicates, *Negative Control*, Positive Control-1, and *Positive Control-2*, which are also technical replicates. The *Positive Control* sample is when MT1 is not present. High output is expected as a function of blue or the output plasmid transfection marker, similar to in Transfection A. *Negative Control* test when MT1 is present but not the output plasmid, note that, similar to Transfection A, TagBFP still gets transfected into this sample to create a heatmap. Uniformly low output is expected for the *Negative Control*. High output is expected for the *Full* at high blue and low green or high plasmid copy number and low MT1 repressor.

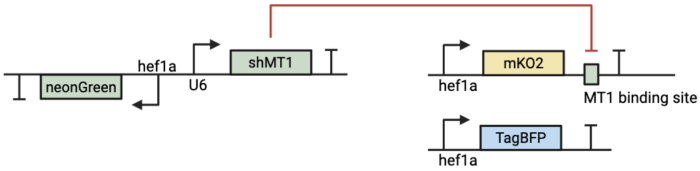


Figure 9: Plasmid-level circuit diagram for Transfection B testing the repression mechanism of MT1 on mKO2 output plasmid with MT1 binding site.

The transfection process for Transfection B also didn't go as expected. First, there is a presence of red fluorescence in the Negative Controls, as seen in Figure 10, even though output plasmids were not introduced into the samples. This could be due to small contamination or fluorescence bleed-through. Also, the *Positive Controls* have significantly smaller blue fluorescence than the Full and *Negative Control*. Also, red fluorescence is very low across all samples. This would yield a low output signal for the *Positive Control*.

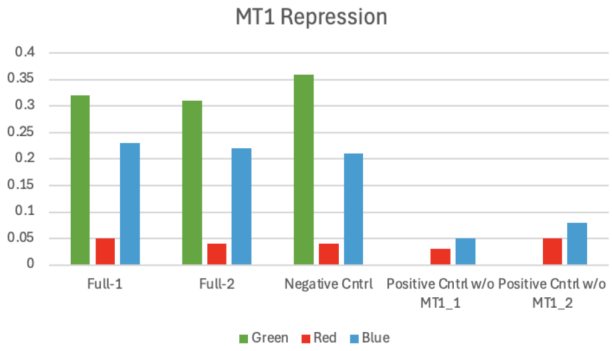


Figure 10: Transfection efficiency for Transfection B for *Full*, *Negative Control*, and *Positive Control* samples.

3.2.1 Gro Simulation 2: Characterizing MT1 Repression

Much similar to simulation 1, Gro simulation 2 seeks to confirm the repression mechanism, but this time for the MT1 miRNA, and confirm the hypothesis that output expression is high at high copy number and low MT1 presence.

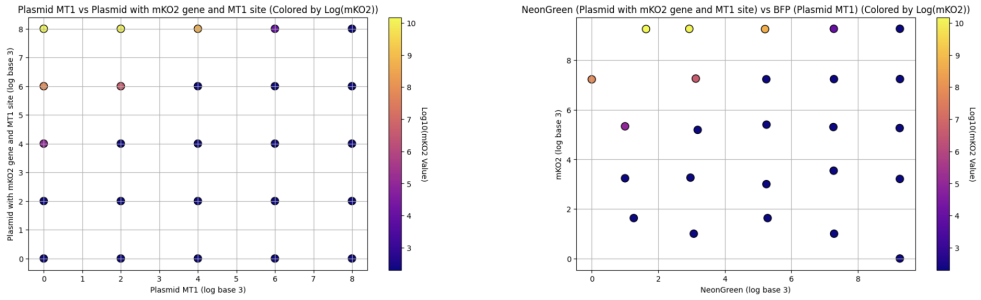


Figure 11: Heatmaps from Gro Simulation 2 to be compared with the *Full* sample from Transfection B. Left and right subplots show the output signal by direct number of plasmids and fluorescence value of transfection markers, respectively.

Figure 11 shows the result from the simulation. Once again, the left plot shows output as a function of direct measurement of plasmid numbers, and the right subplot shows the same output as a function of fluorescence markers, respectively. As seen from the heatmaps, the highest output signal is at high copy numbers (y-axis) and low MT1 presence (x-axis). We expect Transfection B to qualitatively follow the trend in simulation 2. Transfection B showed mixed results, possibly once again due to faulty transfection, but the experimental result does show certain similarity to the simulation result. Detailed discussion of the experimental result is in the following section on the *Full* sample. Note that we observe stronger repression from MT1 compared to CasE. This is due to a combination of factors, such as how MT1 was modeled to be multiple use and CasE was modeled to be single-use. MT1 repression was also modeled with Hill function kinetics, while CasE was modeled to have

linear repression strength with concentration.

3.2.2 Transfection B Results

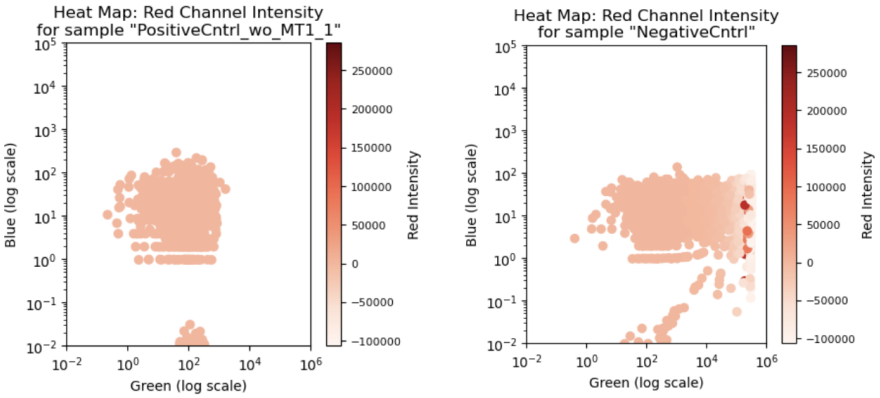


Figure 12: Heatmaps for the *Positive Control* and *Negative Control* in Transfection B.

High output as a function of blue is expected for Positive Control due to the lack of repression of MT1. However, due to the unexpectedly low transfection of the output plasmid and the blue marker *Positive Control* shows an overall lower output signal than expected, as seen in Figure 12. Negative Control has low output as expected, except what seems to be a small second distribution at high green value that shows a high output signal. This is most likely a result of contamination.

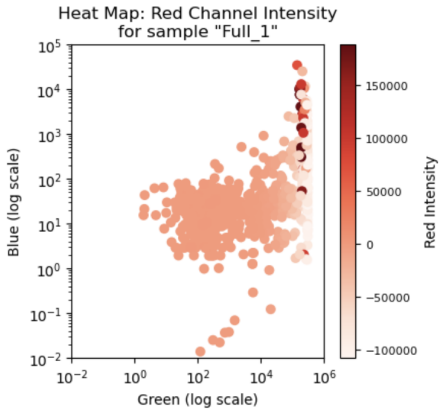


Figure 13: Heatmap for the *Full* in Transfection B.

As seen in Figure 13, the result from the *Full* sample is mixed. We expected a high output signal at high blue and low green value, corresponding to high output plasmid copy number and low MT1 repressor. The heatmap shows the highest output at high blue and high green. This result would suggest that the transfection works as there is a relationship between output and transfected plasmids, but the repression activity of MT1 was not as expected. It

is unclear what could cause this issue, but it is most likely some form of human error, such as faults in the transfection process or during data processing.

3.3 Transfection C - Characterizing MT1 with CasE Activity

Following the characterization of CasE and MT1 repression in Transfection A and B, Transfection C aims to characterize the cascading manner of how MT1 represses CasE, which represses the output expression. Figure 14 shows the plasmid-level circuit diagram for Transfection C. Green is the marker for MT1, blue is the marker for the output plasmid, and red is the output. For Transfection C, we also introduce an infrared signal with iRFP720 as the fourth marker to mark CasE transfection. Figure 15 shows the transfection efficiency for the three samples: *Positive Control*, *Low*, and *Full*. *Positive Control* sample test when MT1 is present but CasE is not present. Because the output plasmid in Transfection C has a CasE binding site, it is expected that *Positive Sample* has a high output signal as a function of the output plasmid transfection market. *Low* sample test when MT1 is not present and CasE is. As CasE expression is not regulated, it is expected that the *Low* sample shows high signal at high blue and low iRed, corresponding to high output plasmid transfection and low CasE. *Full* sample is when both MT1 and CasE are transfected and tests the cascading repression effect on the output. As we now have four degrees of freedom, *Full* sample result needs to be plotted on a 3D heatmap. *Full* sample is expected to have high output at high green and low infrared signal, corresponding to high MT1 and low CasE. This is because high MT1 will repress CasE expression. Lack of CasE will allow for high expression of the output.

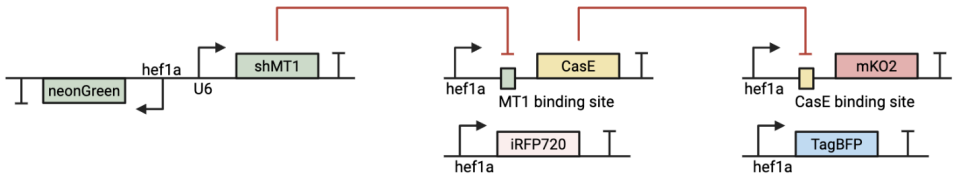


Figure 14: Plasmid-level circuit diagram for Transfection C testing the cascading repression mechanism of MT1 on CasE on the output plasmid with CasE binding site.

As seen in Figure 15, transfection efficiency for Transfection C shows that the transfection process went well. The difference between green signal, which is the maker for MT1, for *Low* and *Full* is, for some reason, very slight but present nonetheless, and iRed signal, which is the marker for CasE, is lower on the no CasE sample compared to the rest.
As seen in Figure 16, the *Positive Control* heatmap result follows expectation such that the output signal increases with a higher blue signal, which is the marker for the transfected output plasmid. Secondly, the red signal stays invariant to iRFP, the marker for CasE, since CasE was not introduced into the sample. The *Low* sample with only CasE and not MT1 also behaved as expected. The heatmap shows high output at high blue and low infrared values, corresponding to high output plasmid copy number and low transfected CasE. Because MT1 was not transfected into the sample, output is expressed more when there's low CasE.

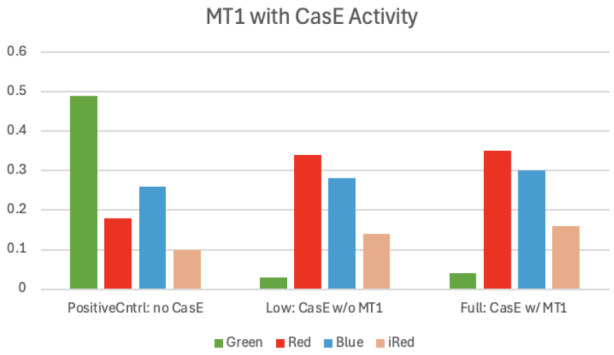


Figure 15: Transfection efficiency for Transfection C for *Positive Control*, *Low*, and *Full* samples.

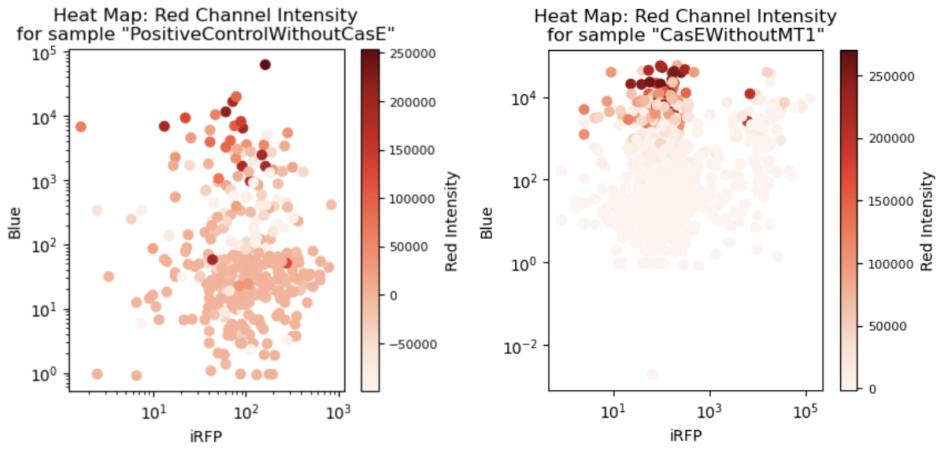


Figure 16: Heatmap for the *Positive Control* and *Low* samples in Transfection C.

Finally, the Full sample of Transfection C tests a system where both CasE and MT1 repressors are present and how they influence the output expression. As seen in Figure 17, the circuit performed as expected. There is a higher expression at high MT1 (infrared), low CasE (green), and high output plasmid (blue). This confirms that the lower branch of the circuit behaves correctly, such that CasE represses the output expression and MT1 represses the expression of CasE. Interestingly, there's also another population that exhibits a high red signal at low green and iRFP region, corresponding to both low MT1 and low CasE plasmids. The second population is "flat" against the infrared-blue plane, which could be the result of certain populations of cells only taking up two of the three plasmids, which could be from contamination or other forms of errors. The population that is not a flat projection on the plane takes up all three plasmids and is the one that exhibits the expected results. It is fair to assume that it is the population of interest and that Transfection C behaved as expected.

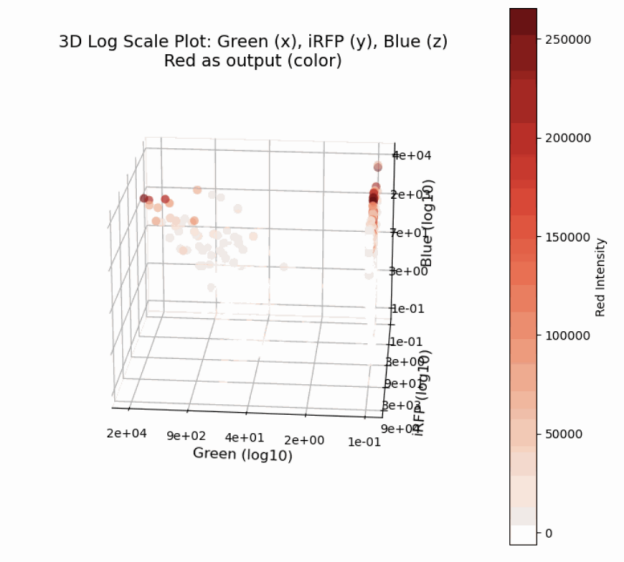


Figure 17: Heatmap of the *Full* sample in Transfection C.

3.4 Transfection D - Characterizing IL6/ADAR Induction

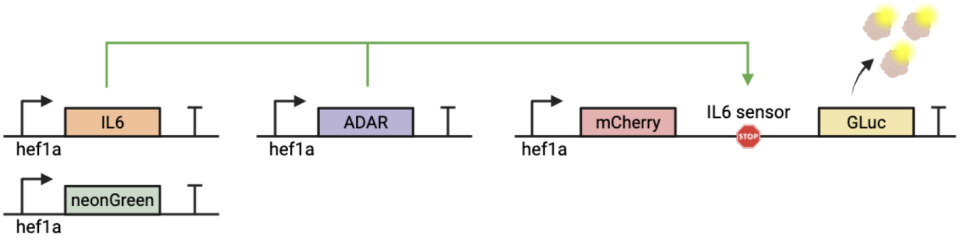


Figure 18: Plasmid-level circuit diagram for Transfection D testing the induction mechanism of IL6 and ADAR on the output plasmid with an IL6 binding site.

Transfections A, B, and C were done to characterize the lower branch of the AND gate that involves the MT1 miRNA. Following that, Transfection D aims to characterize the upper branch of the AND gate that has the IL6 mRNA as input. Figure 18 shows the plasmid-level circuit diagram for Transfection D. Green is a marker for IL6, and red is a marker for the ADAR plasmid. As seen from the diagram. When IL6 is present, it binds to the IL6 sensor site on the output plasmid. When ADAR sees this binding action, it cleaves the output plasmid and inverts the stop codon and allowing for the promotion of the Gaussia luciferase (GLuc) Assay. It is expected that GLuc output is high when both IL6 and ADAR are high. Figure 19 shows the transfection efficiency for Transfection D. The x-axis represents different ratios of IL6 to ADAR, where ADAR is kept constant. As expected, transfecting greater levels of IL6 plasmid was correlated with higher transfection efficiency and higher copy number in cells. However, since the amount of ADAR sensors stays constant over different conditions, the expected result is that the red value on transfection efficiency stays the same. However, Figure 19 shows the increasing level of ADAR as we increased the amount of IL6.

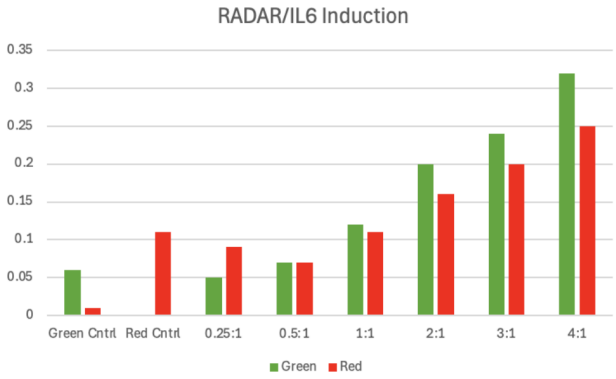


Figure 19: Transfection efficiency for Transfection D for Green Control, Red Control, 0.25:1, 0.5:1, 1:1, 2:1, 3:1, 4:1 IL6 to ADAR ratios samples.

A possible explanation for why red signal increases might be that bleedthrough of green into the red channel contributes to the fluorescence despite constant levels of the ADAR gene. As seen on the green control sample, a small bleedthrough from green to red channel can be seen.

3.4.1 Gro Simulation 3: Characterizing IL6/ADAR Induction

Gro simulating 3 seeks to test IL6 and ADAR induction behavior on the output and confirm the hypothesis that output expression is higher at high copy number, high IL6 to bind to the output plasmid, and high ADAR to convert the stop codon once IL6 binds and promote expression of the output. Figure 20 shows the result from the simulation. The Gro simulation result follows expectations. As high IL6 and high ADAR lead to high expression of output, the heatmaps show higher output signals in the top right corner. Another interesting result from the simulation is that as we simulated the induction relationship at various levels of output plasmid, we found that the induction effect becomes more and more prominent as a function of increasing level of output plasmid present in the cell. As seen in Figure 20, as we move towards the right subfigures, towards higher output copy number, the induction effect becomes more and more localized towards the top right corner of the plot. Unfortunately, Transfection D did not yield any useful results, and so we were not able to confirm the experimental results against the Gro simulation 3 results.

3.4.2 Transfection D Results

As seen from Figure 21, the bottom four rows highlighted in red were used for FACs analysis to determine if the transfection process went well. The top four rows highlighted in green were used for Luciferase assay measurement. As seen from Figure 22, the plate luminescence data were very noisy. Even though every row is are technical replicate, the luminance values vary across all rows. There are also no apparent trends to draw any conclusion from. We would have to repeat the experiment, maybe with a constitutively expressed hef1a promoter on GLuc, to see if GLuc expresses at all in this system. This, unfortunately, means that the upper mRNA branch of the full circuit was not verified.

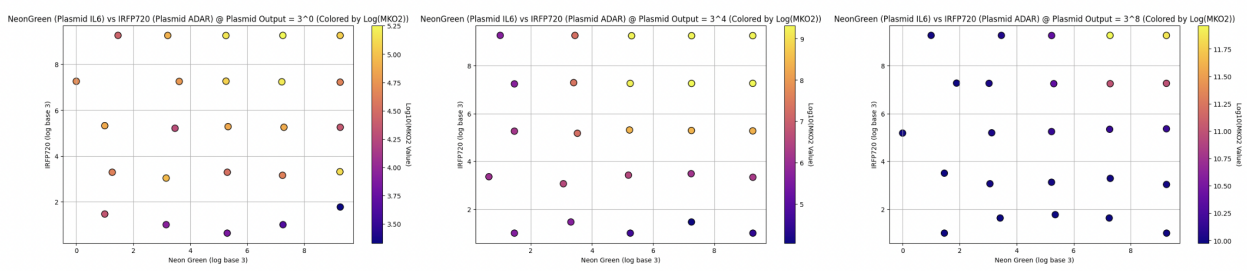


Figure 20: Heatmaps from Gro Simulation 3. All three subplots are from three separate instances of the simulation and show the output signal by fluorescence value of transfection markers at various amounts of output plasmids 1, 81, and 6561, respectively.

	1	2	3	4	5	6	7	8	9	10	11	12
A	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
B	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
C	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
D	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
E	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
F	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
G	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		
H	Blank	Red	Green	0.25:1	0.5:1	1:1	2:1	3:1	4:1	WT		

Figure 21: Well placements for Transfection D.

	1	2	3	4	5	6	7	8	9	10	11	12
A	7	4	4	3	6	9	4	10	3	4	8	6
B	4	6	6	9	4	5	5	6	5	9	4	4
C	21	4	5	4	4	10	6	5	5	4	5	6
D	4	4	7	4	3	6	5	6	6	6	5	4
E	5	5	5	7	4	6	5	5	5	6	5	9
F	6	5	5	4	3	7	4	4	7	5	4	3
G	4	7	5	5	4	6	3	4	6	7	5	4
H	4	4	6	9	5	6	5	3	3	5	4	5

Figure 22: Measured luminescence strength for samples in Transfection D.

3.5 Conclusion

This study demonstrated significant progress toward the design and characterization of a dual-input RNA-based AND gate integrating miRNA (MT1) and mRNA (IL6) regulatory mechanisms. Through a modular experimental approach and computational simulations, individual components of the circuit were evaluated, with several modules confirming the expected behaviors of CasE-mediated repression, MT1 inhibition of CasE, the cascading interactions between MT1 and CasE on output, and the induction mechanism of IL6 and ADAR. The experimental outcomes, although affected by transfection inefficiencies and procedural limitations, broadly aligned with computational predictions generated by Gro simulations.

However, critical technical challenges limited the conclusive verification of the entire synthetic logic circuit. Specifically, the inability to successfully construct and test the integrated output plasmid prevented complete validation of the dual-input logic function. Additionally, inconsistent results in transfection efficiency and unexpected experimental anomalies in certain submodules highlighted the necessity for procedural refinement and replication in future experiments. Despite these obstacles, the foundational work presented herein substantially advances the development of RNA-based logic gates for precise biological control and diagnostics. Future investigations should focus on optimizing plasmid assembly and transfection protocols, rigorously validating the upper IL6/ADAR branch, and systematically addressing observed anomalies through iterative testing and computational modeling. Ultimately, the successful implementation of such synthetic biological circuits could pave the way for novel diagnostic and therapeutic applications, particularly in the context of diseases characterized by distinctive molecular signatures, such as Multiple Sclerosis.

4 Supplementary Material

Table 1: Table of provided level 0 plasmids and descriptions

Plasmid	**Name**	**Description (L1 Carb / L2 Spec)**
pTM01	L1-ST1-2	L1 Backbone
pTM02	pL0-I-E2	E2 Insulator
pTM03	pL0-P-Hef1A	Hef1A promoter
pTM07	pL0-5-Blank	Inert 5' UTR
pTM09	pL0-G-mKO2	mKO2 gene
pTM15	pL0-3-Blank	Inert 3' UTR
pTM16	pL0-T-WPRE	PolyA
DM04	pL0-5-casErec	5'UTR
DM09	CMV_ADAR	ADAR Gene
DM13	pL0-T_Rb Glob pA	Terminator
DM31	plvl0-5-4xFF4	5'UTR
DM104	pLO-5-mT1	5'UTR
DM106	pLO-G-IL6	IL6 Gene
DM107	pLO-G-TagBFP_RADARIL6Sensor_mKO2	Output Gene
CMV	CMV	CMV promoter

Table 2: Table of provided level 1 plasmids

uniqueID	**Plasmid**
hEf1a_CasErec-mKO2-FF4	DM34
hEF1a_CasE	DM43
hEF1a_TagBFP	TM22
hEF1a_neonGreen	TM20
hEF1a_iRFP720	DM33
SB_(P1rev)U6_MT1_(P2)hef1a_neonGreen	DM22
RADARS_Luciferase	ADDGENE182540

Table 3: Table of successfully cloned level 1 plasmids and their level 0 constituents

uniqueID	**Backbone**	**Insulator**	**Promoter**	**5'**	**Gene**	**3'**	**Terminator**
hEF1a_IL6	pTM01	pTM02	TM03	pTM07	DM106	pTM15	pTM16
hEF1a_ADAR	pTM01	pTM02	CMV	pTM07	DM09	pTM15	pTM16
hEF1a_FF4-CasE-mT1	TM01	TM03	DM31	DM09	DM104	DM13	TM02
hEF1a_FF4-mKO2-mT1	TM01	TM03	DM31	TM09	DM104	DM13	TM02
TagBFP/mkO2 output	pTM01	pTM02	TM03	DM04	DM107	pTM15	pTM16

Table 4: Table Summary of Species Functional Descriptions

Species	**Types**	**Descriptions**
mKO2	Output	Output
TagBFP	Transfection Marker	Proxy for copy numbers (depending on specific transfection submodules)
iRFP	Transfection Marker	Proxy for copy numbers (depending on specific transfection submodules)
neonGreen	Transfection Marker	Proxy for copy numbers (depending on specific transfection submodules)
IL6	Active Species	Proxy for general mRNA to be replaced with application-specific mRNAs
ADAR	Active Species	Agent that invert the stop codon on the output plasmid when both mRNA and miRNA are present
MT1	Active Species	Proxy for general miRNA species to be replaced with application-specific miRNAs
CasE	Active Species	Additional logic inverter on miRNA branch as miRNA is intrinsically repressive.

1. Remove multiwell plates containing mammalian cells from the incubator. Make certain that the plates are compatible with the type of luminometer being used.

2. Measuring firefly luciferase activity: Add a volume of Dual-Glo® Luciferase Reagent equal to the culture medium volume to each well and mix. For 96-well plates, typically 75µl of reagent is added to cells grown in 75µl of medium.

3. Wait at least 10 minutes, then measure the firefly luminescence (consult your luminometer manual for proper use of the instrument). Optimal results will be generated if the luminescence is measured within 2 hours of addition of Dual-Glo® Luciferase Reagent.

4. Measuring Renilla luciferase activity: Add a volume of Dual-Glo® Stop & Glo® Reagent equal to the original culture medium volume to each well and mix. As noted in Step 2, this volume is typically 75µl for 96-well plates and 20µl for 384-well plates. Note: Dual-Glo® Stop & Glo® Reagent should be added to plate wells within 4 hours of addition of Dual-Glo® Luciferase Reagent.

5. Wait at least 10 minutes, then measure luminescence. Renilla luminescence should be measured in the same plate order as the firefly luminescence was measured (Step 3). For optimal results, measure luminescence within 2 hours of addition of Dual-Glo® Stop & Glo® Reagent.

Figure 23: Luciferase Assay Protocol Used to Measure Transfection D Results